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COURSE PROJECT REPORT

Course: Trustworthy AI

SPURIOUS CORRELATIONS AND INVARIANT LEARNING

ERM vs IRM vs REx on Colored-MNIST

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ABSTRACT

Machine learning models can perform well on in-distribution data but fail under distribution shift when they rely on spurious correlations. This report studies this problem using Colored-MNIST, where digit shape is the stable signal and color is spuriously correlated with the label during training but largely reversed at test time. We compare ERM, IRMv1, and V-REx under the same training and evaluation pipeline using multiple random seeds and train-domain validation as the primary model-selection protocol. Additional experiments examine the effects of penalty strength, backbone architecture, number of environments, additional spurious features, sample size, and dataset choice. The results show that ERM performs well on the training environments but degrades substantially under the OOD setting. IRMv1 is strongly affected by model selection and also shows considerable variation across random seeds. V-REx provides more stable behavior across the tested penalty settings. The ablation experiments further show that invariant learning becomes more difficult when the number of environments is reduced or additional spurious features are introduced. Diagnostic analyses using calibration, Integrated Gradients, and linear probes show that reduced reliance on color does not necessarily mean that color information is removed from the learned representation. Overall, the results highlight the importance of model selection, repeated runs, and diagnostic analysis when evaluating OOD generalization under spurious correlations.

Keywords: spurious correlation, invariant learning, IRM, V-REx, OOD generalization, model selection, trustworthy AI.

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1. Introduction

Some machine learning models achieve high accuracy on in-distribution data; however, they may fail under distribution shift, resulting in poor performance in practical applications. One reason for this problem is that some features are correlated with the target label in the training data, but this correlation does not hold in real-world settings. This is called a *spurious correlation*. A common example of this phenomenon is the cow-camel classification problem, where cows are predominantly observed on grassy backgrounds while camels are associated with sandy backgrounds in the training data. As a result, a classifier may learn to rely on the background rather than the characteristics of the animal itself, leading to poor performance when this correlation changes at test time [1].

To study this problem, we set up a controlled environment using **Colored-MNIST** [2], where digit shape serves as the stable signal, while color is spuriously correlated with the label during training and anti-correlated at test time. This setting provides a simple way to observe how models behave when a spurious correlation is present and how different training objectives respond to changes in the data distribution. Based on this setup, the report focuses on the following research questions:

- **RQ1 (reproduce):** Does ERM rely on color and experience a drop in OOD performance, and how do invariant learning objectives behave under the same setting?
- **RQ2 (break):** Under what conditions does invariant learning become less effective, considering penalty strength, backbone capacity, number of environments, a second spurious feature, sample size, and dataset choice?
- **RQ3 (diagnose):** Which diagnostics can help indicate whether a model relies less on color, including calibration, attribution methods, and representation probes?

The report includes three main parts:

1. We evaluate ERM, IRMv1, and V-REx across three random seeds and report the mean$\pm$std results under two model-selection protocols: train-domain validation and OOD oracle [3].
2. We conduct 18 ablation settings covering the IRM-λ sweep, REx-β sweep, MLP and CNN backbones, single-environment training, a second spurious patch, small-n settings, and transfer to Colored-FashionMNIST
3. We examine model behavior using reliability analysis, Integrated Gradients, and linear probes, with particular attention to the Reliability and Explainability aspects of trustworthy AI

2. Related work

2.1. Invariant Risk Minimization (IRM)

Arjovsky et al. [2] introduce Invariant Risk Minimization (IRM), which aims to learn a representation Φ such that the same classifier w performs well across all training environments. In practice, the commonly used IRMv1 objective adds a gradient-based penalty to encourage this invariance (see §3). Later studies show that IRMv1 can be sensitive to the random seed and the penalty weight λ. Several extensions have therefore been proposed, including game-based optimization [4], gradient-variance regularization such as Fishr [5], and sparse variants of IRM [6].

2.2. Risk Extrapolation (REx) and Group DRO

V-REx [7] encourages similar risks across training environments by penalizing the variance of environment-wise risks. This provides a simpler alternative to the IRMv1 penalty. Group DRO [8] takes a different approach by minimizing the loss of the worst-performing environment. In this report, a simplified Group DRO objective is included as an additional reference method.

2.3. Limitations of invariant learning

Previous work also studies cases where invariant learning may not work as expected. Rosenfeld et al. [9] show that IRM requires enough training environments relative to the number of spurious features, and that learning from a single environment is not sufficient in their setting. Zhou et al. [6] further show that over-parameterized models can make invariant learning more difficult. These results motivate several of the stress tests used in our experiments.

2.4. Model selection and diagnostic methods

Model selection is also important when evaluating OOD generalization. DomainBed [3] shows that reported performance can depend strongly on how models are selected, especially when comparing train-domain validation with OOD-based selection. In addition, DFR [10] and JTT [11] suggest that useful features may already be present in the learned representation even when the final classifier does not use them effectively. This motivates the use of linear probes in our analysis. For explainability, we use Integrated Gradients [12] as the main attribution method, while vanilla saliency is kept only as a simple baseline because of known limitations [13].

3. Problem formulation and methods

3.1. Structural causal model

Let $D \in \{0, \dots, 9\}$ denote the original MNIST digit. We first define the binary label $\tilde{Y} = \mathbf{1}[D \geq 5]$, which separates digits 0–4 from digits 5–9. To introduce label noise, the observed label is generated as $Y = \text{flip}(\tilde{Y}; 0.25)$, meaning that the binary label is flipped with probability 0.25.

The color variable C is then generated from the observed label as $C = \text{flip}(Y; p_e)$, where p_e depends on the environment e. The final image $X \in \mathbb{R}^{3\times28\times28}$ contains the original digit shape rendered in red when $C = 0$ and in green when $C = 1$.

This construction separates two sources of information. The digit shape is determined by D and remains unchanged across environments, so it acts as the stable signal. In contrast, the relationship between color C and the observed label Y depends on p_e, so color acts as the spurious signal. By changing p_e between training and test environments, we can directly study whether a model relies on the stable signal or on the easier but environment-dependent color signal.

3.2. Input

The input to the model is a colored digit image $x \in \mathbb{R}^{3\times28\times28}$. Each input contains both the original digit shape and the assigned color. The shape carries information about the underlying digit D, while the color is introduced according to the environment-specific variable C.

Because both signals are available in the same image, the model can potentially solve the task using either of them. However, only the digit shape remains reliable when the data distribution changes across environments.

3.3. Output

The task is binary classification. Given an input image x, the model estimates the probability

$$P(Y = 1 \mid X = x),$$

and produces a prediction $\hat{y} \in \{0, 1\}$.

The target label is derived from the original digit group: digits 0–4 correspond to $\tilde{Y} = 0$, while digits 5–9 correspond to $\tilde{Y} = 1$. The observed label Y is obtained after applying the 25% label noise described above. Therefore, even a classifier that perfectly recovers the digit group cannot achieve perfect accuracy with respect to Y.

3.4. Environments

We use two training environments and one OOD test environment:

$$e_1 : p_e = 0.1, \qquad e_2 : p_e = 0.2, \qquad e_{\text{te}} : p_e = 0.9.$$

The parameter p_e controls the probability that the color is flipped relative to the observed label. When p_e is small, color is strongly correlated with the label. Therefore, in the two training environments, color provides an easy predictive signal. In the OOD test environment, however, $p_e = 0.9$, so this relationship is almost reversed.

This shift creates the main difficulty of the task. A classifier that mainly uses color can perform well on the training distribution but fail when evaluated on the OOD environment. In contrast, a classifier that relies on digit shape is unaffected by the change in color correlation.

Under this construction, a color-only rule achieves approximately 0.85 ID accuracy but only around 0.10 OOD accuracy. A shape-only oracle achieves approximately 0.75 accuracy across all environments, with this upper bound caused by the 25% label noise.

3.5. Learning objectives

For each environment e, let

$$R^e(\theta) = \mathbb{E}_{(x,y)\sim P_e}\left[\ell(f_\theta(x), y)\right],$$

where ℓ denotes the binary cross-entropy loss and $R^e(\theta)$ is the risk of model f_θ in environment e. The methods considered in this report use the same environment-specific risks but optimize them using different objectives.

3.5.1. Empirical Risk Minimization (ERM)

ERM minimizes the average risk across all training environments:

$$\min_\theta \frac{1}{E} \sum_e R^e(\theta).$$

ERM does not explicitly distinguish between stable and spurious signals. Therefore, if color provides an easier predictive signal during training, the model may rely on color even when this relationship does not remain valid in the OOD environment.

3.5.2. Invariant Risk Minimization (IRMv1)

IRM aims to learn a representation for which the same classifier remains suitable across different training environments. In practice, IRMv1 introduces a gradient-based penalty:

$$\min_\Phi \sum_e R^e(w \circ \Phi) + \lambda \sum_e \left\|\nabla_{w=1} R^e(w \cdot \Phi)\right\|^2.$$

The penalty encourages the learned representation to support a shared classifier across environments. In our experiments, the penalty is activated after an annealing period of 190 optimization steps, so $\lambda = 0$ before step 190.

3.5.3. Variance Risk Extrapolation (V-REx)

V-REx encourages the model to achieve similar risks across training environments. Its objective is defined as

$$\min_{\theta} \operatorname{mean}_e \left(R^e\right) + \beta \cdot \operatorname{Var}_e \left(R^e\right),$$

where β controls the strength of the variance penalty. A larger value of β places more emphasis on reducing differences in risk between environments.

3.5.4. GroupDRO-lite

GroupDRO-lite focuses on the environment with the highest risk and minimizes the worst-environment loss:

$$\min_{\theta} \max_{e} R^e(\theta).$$

Unlike ERM, which optimizes the average risk, this objective gives more attention to the environment in which the model currently performs worst. In this report, GroupDRO-lite is included as an additional reference objective rather than a main method of analysis.

For IRMv1, we use the standard dummy-classifier formulation with $w = 1.0$ to compute the gradient penalty. Evaluation metrics, dataset sizes, and the model-selection protocol used throughout the experiments are described in §4.2.

3.6. Implementation and optimization

3.6.1. Pipeline

The implementation follows a simple `data`→`train`→`eval`→`viz` pipeline.

The data module, `src/cmnist/data.py`, constructs the Colored-MNIST datasets following the protocol introduced by Arjovsky et al. [2]. It also supports Colored-FashionMNIST and an optional 3×3 spurious patch used in several failure-case experiments. Each generated sample is assigned to a predefined environment according to the corresponding color-flip probability.

The training stage receives these environment-specific datasets together with a model architecture and a learning objective. The same pipeline is used across all experiments so that differences in results mainly come from the objective, model architecture, or experimental setting rather than from changes in the training procedure.

3.6.2. Models and optimization

Two backbone architectures are used. The first is a multilayer perceptron (MLP) with two hidden layers of 256 units each. The second is a small convolutional neural network (CNN), which provides a higher-capacity image-specific backbone for comparison.

The learning objectives are implemented in `losses.py` and include ERM, IRMv1, V-REx, and GroupDRO-lite. Training is handled by `train.py` using the Adam optimizer with a learning rate of 10^{-3}, a batch size of 2048, and 2000 optimization steps for the main experiments.

For objectives that use a penalty term, the penalty is activated only after an annealing period of 190 steps. Before this threshold, the penalty weight is set to zero. Therefore, IRMv1 and V-REx initially behave like ERM and only begin enforcing their additional constraints after the annealing stage.

During training, the pipeline records checkpoints according to both ID validation performance and OOD performance. These checkpoints are later used to compare the effect of different model-selection protocols.

3.6.3. Evaluation

The evaluation module, `eval.py`, measures model performance separately for each environment. The reported metrics include per-environment accuracy, worst-group accuracy, the performance gap between environments, and Expected Calibration Error (ECE).

These metrics are intended to capture different aspects of model behavior. Accuracy measures predictive performance, worst-group accuracy reflects the weakest environment, the environment gap indicates how much performance changes under distribution shift, and ECE provides information about calibration.

Additional diagnostic outputs, including attribution maps and linear-probe results, are produced for the analysis in RQ3. Visualization utilities are separated from the training and evaluation code so that the same trained checkpoints can be reused for different diagnostic analyses.

3.6.4. Experiment budget

To manage the experimental workload, we use three execution modes.

- `fast`: 500 training steps and one random seed, used mainly to verify that the full pipeline runs correctly.
- `base`: 2000 training steps for ERM, IRMv1, and V-REx across three random seeds. This configuration is used for the main comparison in RQ1.
- `heavy`: extended experiments including penalty sweeps, backbone comparisons, failure-case variants, and transfer to Colored-FashionMNIST. This configuration contains 54 runs in total.

All main experiments use random seeds $\{0, 1, 2\}$. Deterministic cuDNN settings are enabled to reduce variation caused by non-deterministic GPU operations. The full experimental suite requires approximately three hours of GPU time on a Colab T4 with 16 GB of memory.

4. Experimental setup

4.1. Purpose of evaluation

The experimental evaluation is designed around the three research questions introduced in §1. For **RQ1 (reproduce)**, we examine how ERM behaves under distribution shift and compare it with invariant learning objectives under the same model-selection protocol. For **RQ2 (break)**, we study how the behavior of invariant learning changes when varying the penalty strength, backbone capacity, number of environments, number of spurious features, sample size, and dataset. For **RQ3 (diagnose)**, we use calibration, attribution methods, and representation probes to investigate whether the model reduces its reliance on the spurious feature.

Section 5 presents the results according to these three research questions.

4.2. Evaluation protocol

4.2.1. Data

All experiments follow the Colored-MNIST construction described in §3. The original MNIST training set is divided into two disjoint training environments, each containing approximately 30,000 images. The first environment uses $p_e = 0.1$, while the second uses $p_e = 0.2$. The full MNIST test set, containing 10,000 images, is used as the OOD test environment with $p_e = 0.9$, where the correlation between color and label is largely reversed.

For the small-n experiments, each training environment is reduced to 6,250 samples to examine the effect of limited training data. For the dataset-transfer experiments, the same environment-generation procedure is applied to FashionMNIST, allowing us to observe whether the behavior found on Colored-MNIST also appears on a different image dataset.

4.2.2. Metrics

We report several performance metrics. The average ID accuracy across the two training environments is

$$\text{ID-avg} = \frac{\text{Acc}_{e_1} + \text{Acc}_{e_2}}{2}.$$

OOD accuracy is the classification accuracy on the held-out test environment,

$$\text{OOD accuracy} = \text{Acc}_{e_{\text{te}}}.$$

To measure the change in performance under distribution shift, we also report the OOD gap:

$$\text{OOD gap} = \text{ID-avg} - \text{OOD accuracy}.$$

A larger gap indicates a larger reduction in accuracy when moving from the training environments to the OOD environment.

To examine performance across combinations of label and color, the test samples are divided into groups $g = (y, c) \in \{0, 1\}^2$. We report the worst-group accuracy as

$$\text{Worst-group accuracy} = \min_g \text{Acc}_g.$$

This metric indicates whether good average performance hides particularly poor performance on one label–color subgroup. Finally, model calibration is evaluated using Expected Calibration Error (ECE) with 15 confidence bins.

4.2.3. Model selection

The primary model-selection protocol is **train-domain validation**, where the checkpoint with the highest ID-avg validation accuracy is selected without using information from the OOD test environment. This protocol is used for the main experimental results.

For comparison, we also report an **OOD oracle** result, where the checkpoint is selected directly according to OOD test accuracy. Since this protocol uses information from the test environment, it is not used as the primary evaluation setting; instead, it provides a reference for examining the effect of model-selection strategy.

Unless otherwise stated, all reported results are presented as mean$\pm$standard deviation over three random seeds.

5. Results and discussion

5.1. Main comparison (RQ1)

Table 1 and Fig. 1 summarize the main comparison under the train-domain validation protocol. ERM achieves very high accuracy on the training environments but shows a substantial decrease in OOD accuracy. IRMv1 shows a similar drop, together with noticeably larger variation across random seeds. In contrast, V-REx achieves higher average OOD accuracy and lower seed variance than ERM and IRMv1 under this evaluation setting.

Table 1: Main OOD comparison under train-domain validation (mean±std over 3 seeds).

Method	Train (ID avg)	Test (OOD)	OOD gap	Worst group	ECE
ERM	0.994	0.315 ± 0.010	0.679	0.243 ± 0.010	0.017
IRMv1	0.801	0.243 ± 0.172	0.558	0.010 ± 0.009	0.309
V-REx	0.806	0.424 ± 0.014	0.382	0.339 ± 0.035	0.058

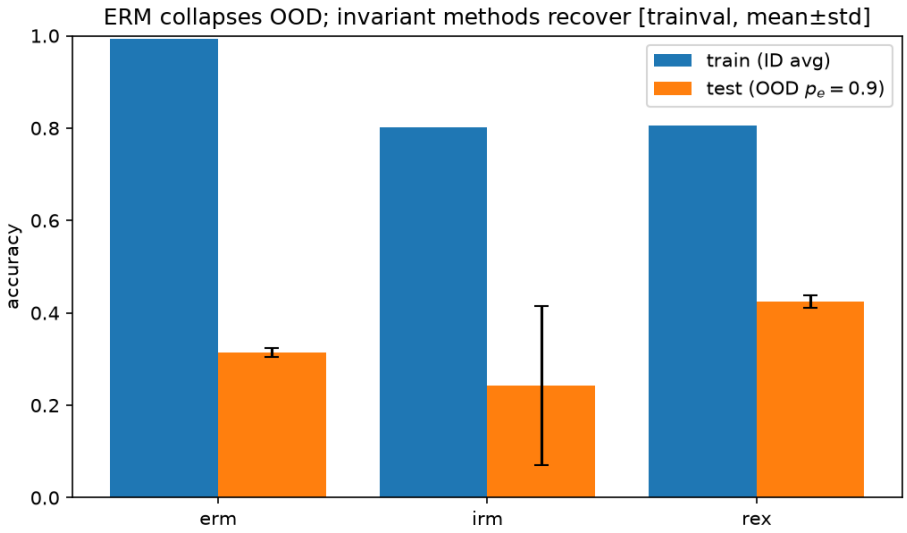


Figure 1: Training and OOD test accuracy under train-domain validation. Error bars indicate the standard deviation over three random seeds.

5.1.1. Effect of model selection

The choice of model-selection protocol also affects the reported OOD performance. Table 2 compares train-domain validation, OOD-oracle selection, and the final checkpoint, following the distinction discussed in DomainBed [3].

Table 2: OOD accuracy under different model-selection protocols (mean±std over 3 seeds).

Method	Train-domain validation	OOD oracle	Last checkpoint
ERM	0.315 ± 0.010	0.331 ± 0.024	0.304 ± 0.014
IRMv1 ($\lambda = 10^4$)	0.243 ± 0.172	0.799 ± 0.030	0.754 ± 0.109
V-REx ($\beta = 10^3$)	0.424 ± 0.014	0.563 ± 0.045	0.439 ± 0.028

ERM shows relatively similar OOD accuracy across the three protocols, while V-REx shows a moderate improvement under OOD-oracle selection. IRMv1 is much more sensitive to the selection strategy: its mean OOD accuracy increases from 0.243 ± 0.172 under train-domain validation to 0.799 ± 0.030 under OOD-oracle selection. The final checkpoint also reaches 0.754 ± 0.109.

These results indicate that IRMv1 can reach checkpoints with high OOD accuracy during training, but such checkpoints are not consistently selected using only train-domain validation. Therefore, the OOD-oracle result is used only as a reference and should not be interpreted as a practical model-selection protocol.

5.2. Penalty stability (RQ2a)

Table 3 summarizes the penalty-strength sweep over three random seeds. For each method and penalty strength, it reports the arithmetic mean and sample standard deviation ($n - 1$ denominator).

Table 3: Penalty-strength sweep under train-domain validation (mean±std over 3 seeds).

Method	Penalty	Train	Test OOD	Gap	Worst	ECE
IRMv1	$\lambda = 100$	0.820	0.365 ± 0.090	0.455	0.253 ± 0.164	0.137
IRMv1	$\lambda = 1000$	0.799	0.286 ± 0.255	0.514	0.130 ± 0.187	0.399
IRMv1	$\lambda = 10000$	0.798	0.286 ± 0.255	0.512	0.127 ± 0.184	0.404
IRMv1	$\lambda = 30000$	0.800	0.281 ± 0.247	0.519	0.118 ± 0.167	0.407
V-REx	$\beta = 100$	0.940	0.293 ± 0.032	0.647	0.194 ± 0.077	0.064
V-REx	$\beta = 1000$	0.804	0.437 ± 0.033	0.368	0.343 ± 0.032	0.077
V-REx	$\beta = 10000$	0.785	0.457 ± 0.001	0.328	0.388 ± 0.021	0.087

Table 4 reports the individual OOD and worst-group accuracies underlying these averages. Each row corresponds to one method and penalty strength; its three seed columns are separate training runs.

Table 4: OOD and worst-group accuracy across seeds and penalty strengths.

Method	Penalty	Seed 0 OOD / Worst	Seed 1 OOD / Worst	Seed 2 OOD / Worst
IRMv1	$\lambda = 100$	0.4369 / 0.3853	0.3940 / 0.3029	0.2646 / 0.0695
IRMv1	$\lambda = 1000$	0.1104 / 0.0000	0.5786 / 0.3436	0.1678 / 0.0449
IRMv1	$\lambda = 10000$	0.1083 / 0.0000	0.5782 / 0.3383	0.1703 / 0.0425
IRMv1	$\lambda = 30000$	0.1084 / 0.0000	0.5642 / 0.3094	0.1705 / 0.0443
V-REx	$\beta = 100$	0.3118 / 0.2390	0.3105 / 0.2383	0.2565 / 0.1057
V-REx	$\beta = 1000$	0.4008 / 0.3148	0.4643 / 0.3370	0.4451 / 0.3787
V-REx	$\beta = 10000$	0.4574 / 0.3975	0.4570 / 0.3642	0.4555 / 0.4022

For IRMv1, OOD accuracy changes relatively little within each seed when λ increases from 10^3 to 3×10^4, whereas differences between seeds are large. Over this range, seed 0 achieves OOD accuracy of approximately 0.108–0.110, seed 1 achieves 0.564–0.579, and seed 2 achieves 0.168–0.171. Seed 0 has zero worst-group accuracy at all three of these penalty strengths; seed 2 has low but nonzero worst-group accuracy (0.042–0.045). The $\lambda =$

10^2 setting behaves differently, with OOD accuracies of 0.437, 0.394, and 0.265 for seeds 0, 1, and 2, respectively. Thus, the limited sensitivity to penalty strength applies mainly to the higher-penalty range, rather than to the entire sweep. These results establish substantial variation between runs, but do not isolate initialization or early training dynamics as its cause.

For V-REx, mean OOD accuracy increases from 0.293 at $\beta = 10^2$ to 0.437 at $\beta = 10^3$ and 0.457 at $\beta = 10^4$. OOD standard deviation does not decrease monotonically: it is 0.032, 0.033, and approximately 0.001, respectively. At $\beta = 10^4$, the individual OOD accuracies are 0.4574, 0.4570, and 0.4555, showing close agreement across the tested seeds. Worst-group accuracy remains more variable (0.388 ± 0.021). The highest tested V-REx penalty therefore gives more consistent OOD accuracy, although consistency alone does not imply high predictive performance. Together, these results support reporting both aggregate and individual-seed results.

5.3. Backbones, failure cases, and transfer (RQ2b)

These experiments examine how changes to model architecture, training environments, and data affect the behavior observed in the main Colored-MNIST experiment. Table 5 first examines backbone architecture and the number of training environments.

Table 5: Backbone and environment ablations under train-domain validation (mean±std over 3 seeds).

Method	Configuration	Train	Test OOD	Gap	Worst	ECE
ERM	MLP	0.992	0.300 ± 0.006	0.692	0.219 ± 0.012	0.048
ERM	CNN	0.934	0.301 ± 0.029	0.633	0.197 ± 0.075	0.045
IRMv1	MLP	0.798	0.286 ± 0.255	0.512	0.127 ± 0.184	0.404
IRMv1	CNN	0.575	0.539 ± 0.084	0.036	0.099 ± 0.146	0.389
IRMv1	Single training environment	0.778	0.431 ± 0.243	—	0.211 ± 0.323	0.165

5.3.1. Backbone architecture

Table 5 compares the MLP used in the main experiments with a small CNN. Under train-domain validation, the CNN achieves an OOD accuracy of 0.539 ± 0.084, compared with 0.286 ± 0.255 for the MLP. The CNN therefore shows both a higher mean OOD accuracy and lower variation in this experiment.

However, its worst-group accuracy remains relatively low, indicating that improved average OOD accuracy does not necessarily correspond to uniformly good performance across all label–color groups. The backbone architecture therefore affects the observed performance, but does not by itself remove the difficulties caused by the spurious correlation.

5.3.2. Single training environment

Table 5 also reports IRMv1 with only one training environment. The resulting OOD accuracy is 0.431 ± 0.243, with a large standard deviation across the three random seeds. Although the mean accuracy is not uniformly low, the large variation indicates that the result is considerably less stable when the model cannot compare patterns across multiple environ-

ments. This setting removes an important source of information used by invariant learning: variation between environments.

Table 6 examines additional spurious features, reduced sample size, and transfer to Colored-FashionMNIST.

Table 6: Spurious-feature, sample-size, and dataset-transfer ablations under train-domain validation (mean±std over 3 seeds).

Method	Configuration	Train	Test OOD	Gap	Worst	ECE
ERM	Additional 3×3 patch	0.998	0.146 ± 0.002	0.853	0.089 ± 0.005	0.020
IRMv1	Additional 3×3 patch	0.909	0.117 ± 0.008	0.791	0.080 ± 0.026	0.325
ERM	$n = 6{,}250$ per environment	1.000	0.289 ± 0.003	0.711	0.212 ± 0.005	0.012
IRMv1	$n = 6{,}250$ per environment	0.756	0.356 ± 0.272	0.400	0.119 ± 0.138	0.306
ERM	Colored-FashionMNIST	0.944	0.260 ± 0.009	0.684	0.161 ± 0.021	0.178
IRMv1	Colored-FashionMNIST	0.600	0.650 ± 0.063	-0.050	0.346 ± 0.148	0.218

5.3.3. Additional spurious feature

We introduce a second spurious feature in the form of a 3×3 patch (Table 6). Under this setting, the OOD accuracy of IRMv1 decreases to 0.117 ± 0.008, which is close to the ERM result of 0.146 ± 0.002. This result is consistent with the theoretical analysis of Rosenfeld et al. [9], which relates the ability to identify invariant features to the number of environments relative to the number of independent spurious features. In this experiment, adding another spurious signal makes the invariant learning problem more difficult without adding another training environment.

5.3.4. Limited training data

The small-n setting in Table 6 reduces each training environment to 6,250 samples. Compared with the standard setting, the results show greater variation across random seeds. This suggests that reducing the amount of training data can make the learned solution less stable. Since both the stable digit-shape signal and the spurious color signal must be estimated from fewer observations, differences caused by initialization and optimization may have a larger effect on the final model.

5.3.5. Transfer to Colored-FashionMNIST

Finally, we repeat the same environment-generation procedure using FashionMNIST instead of MNIST. Under this setting, IRMv1 achieves an OOD accuracy of 0.650 ± 0.063, while ERM obtains 0.260 ± 0.009. This shows that the difference between ERM and IRMv1 observed in Colored-MNIST is not limited to the original MNIST digits under the tested transfer setting.

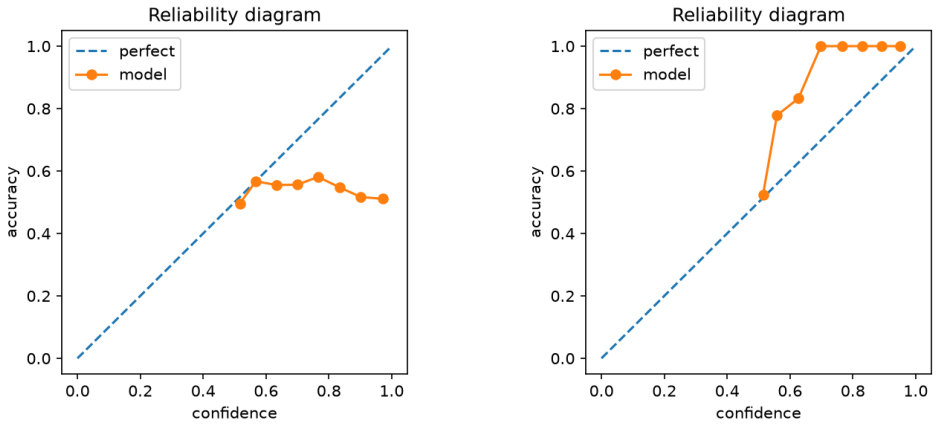
Overall, these experiments show that the behavior of invariant learning depends on several properties of the experimental setting. Additional spurious features, fewer environments, and reduced training data can make the problem less stable, while the choice of backbone can substantially change the observed OOD accuracy. The Colored-FashionMNIST experiment further provides a second dataset on which the same evaluation procedure can be examined.

5.4. Diagnostics (RQ3)

The previous experiments evaluate the models mainly through predictive performance. To better understand how the models use the available information, we also examine their calibration, input attributions, and learned representations.

5.4.1. Calibration

Figure 2 compares the reliability diagrams of ERM and IRM on the OOD test set. ERM obtains an ECE of 0.055, while IRM obtains a higher ECE of 0.231. This indicates that the two models differ not only in OOD accuracy but also in how well their predicted confidence matches the observed correctness.



(a) ERM (OOD, ECE 0.055). (b) IRM (OOD, ECE 0.231).

Figure 2: Reliability diagrams on the OOD test set. The diagonal represents perfect calibration.

5.4.2. Attribution analysis

To inspect which parts of the input contribute to the prediction, we use Integrated Gradients as the main attribution method. Figure 3 shows that the attributions of ERM are more strongly associated with the color-related regions, while IRM places relatively more emphasis on the digit strokes.

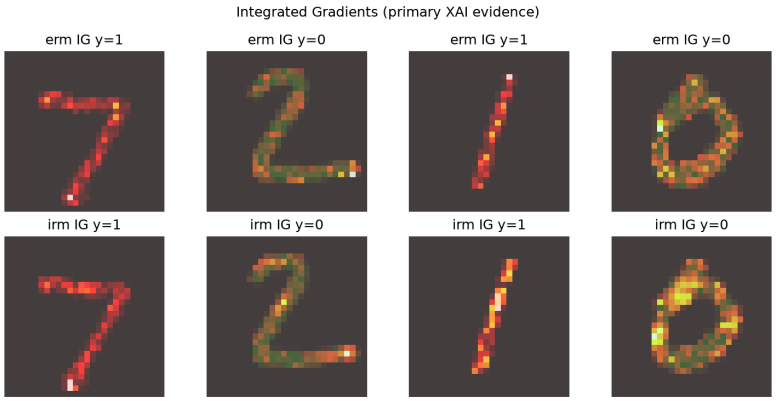


Figure 3: Integrated Gradients for ERM (top row) and IRM (bottom row). ERM shows stronger attribution to color-related regions, while IRM places more emphasis on digit strokes.

For reference, we also report vanilla saliency maps in Fig. 4. However, vanilla saliency is used only as a secondary baseline because gradient-based saliency methods are known to fail several sanity checks [13].

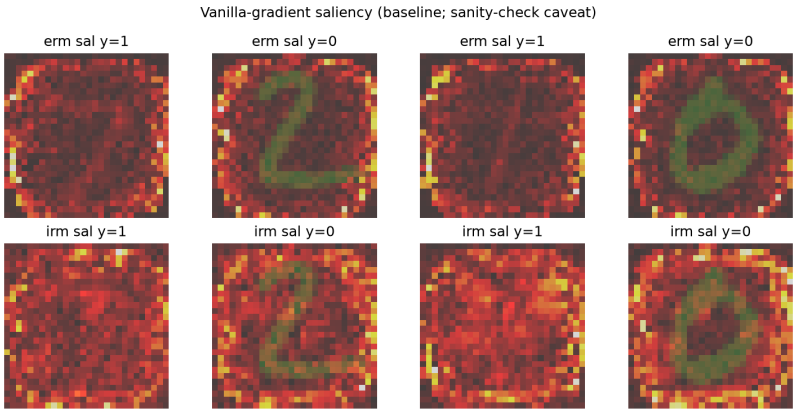


Figure 4: Vanilla saliency maps for ERM and IRM. These results are included as a secondary baseline; Integrated Gradients in Fig. 3 is used as the main attribution analysis.

5.4.3. Representation probe

Finally, we train a linear probe on the penultimate-layer representations to predict the color variable. The probe reaches a held-out accuracy of 1.00 for both ERM and IRM, showing that color information remains fully decodable from the learned representations in both cases.

This result is important when interpreting the attribution results. Although IRM appears to rely less on color for its final prediction, the color information has not been removed from the representation itself. Instead, the difference may occur in how the final classification layer uses the available features. This observation is consistent with previous findings that useful and spurious information can remain in the learned representation while being weighted differently by the final classifier [10].

6. Conclusion

This report examined how spurious correlations affect out-of-distribution (OOD) generalization on Colored-MNIST. ERM, IRMv1, and V-REx were compared using a common pipeline, three random seeds, and complementary diagnostics.

For **RQ1**, high training accuracy did not translate into reliable OOD performance. ERM achieved an OOD accuracy of 0.315 ± 0.010. IRMv1 was particularly sensitive to checkpoint selection: its accuracy increased from 0.243 ± 0.172 under train-domain selection to 0.799 ± 0.030 under OOD-oracle selection. This difference shows that reaching a useful checkpoint and selecting it without OOD information are separate challenges. The oracle result is therefore a diagnostic reference rather than a practical selection rule.

For **RQ2**, increasing the IRMv1 penalty did not consistently improve performance. Results varied substantially between seeds, while changing λ within the higher-penalty range had relatively little effect within each seed. V-REx attained its most consistent OOD result at $\beta = 10^4$ (0.457 ± 0.001), although low variance alone does not imply high accuracy. Backbone architecture, the number of environments, an additional spurious patch, and sample size also affected performance. Colored-FashionMNIST provided a second dataset for testing the same evaluation procedure.

For **RQ3**, attribution and probe analyses distinguished the information encoded in a representation from the information used for prediction. Integrated Gradients suggested greater attention to digit shape for IRM than for ERM, yet a linear color probe achieved held-out accuracy of 1.00 for both models. Reduced apparent reliance on color therefore did not imply that color information had been removed from the representation.

These findings support reporting checkpoint-selection rules, individual-seed results, and diagnostics alongside average OOD accuracy. Penalty strength should be assessed together with architecture and training conditions rather than treated as a sufficient guarantee of invariance.

The conclusions are limited to the tested synthetic shifts in MNIST and FashionMNIST. Real-world datasets such as Waterbirds or CelebA, larger models, and broader hyperparameter searches remain directions for future work. The present results characterize the evaluated settings and do not establish a general ranking of OOD learning methods.

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A. Reproducibility

The source code, experiment configurations, and scripts used in this report are available at:

`https://github.com/namng194/spurious-correlations-invariant-learning`

A.1. Environment setup

The experiments were run with Python 3.12. The environment can be created using:

```
uv venv --python 3.12 .venv
uv sync --extra dev
```

The test suite can be executed with:

```
.venv/bin/python -m pytest tests/ -x -q
```

A.2. Main experiments

The main experiments can be reproduced with:

```
.venv/bin/python scripts/run_sweep.py \
  --config configs/base.yaml \
  --out outputs/sweep
```

The extended ablation experiments can be reproduced with:

```
.venv/bin/python scripts/ablate.py \
  --config configs/heavy.yaml \
  --out outputs/ablate
```

A.3. Figures and diagnostics

The main plots are generated with:

```
.venv/bin/python scripts/make_plots.py \
  --in outputs/sweep/results.json \
  --out outputs/figs
```

Diagnostic figures are generated with:

```
.venv/bin/python scripts/make_diagnostics.py \
  --config configs/fast.yaml \
  --out outputs/figs
```

A.4. Saved results

The raw experimental results are stored in:

- `outputs/sweep/results.json`
- `outputs/ablate/ablate_full.json`

These files contain the values used to generate the tables and figures reported in Section 5.